

Jake Capmeil.

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01 EDUCATION

Hunter College High School

Class of 2028

New York, NY

Relevant coursework: Honors Computer Science, Extended-Honors Math, Honors Physics, AP CS A.

Awards: National French Grand Concours, native-speakers division — two-time Gold Medalist.

02 RESEARCH & ACADEMIC PROGRAMS

Carnegie Mellon University · CS Scholars Program

Summer 2026

Fully funded, merit-based program with a ~4% acceptance rate. Four-week resident taking college-level CS and Graph Theory courses alongside a research project presented in a closing symposium.

Threshold Tuning Under Class Imbalance for Predicting Student Preference in AI-Assisted Tutoring

CMU Research

Modeled student preference for AI versus human tutoring under CMU research mentorship, comparing three classifiers with stratified k-fold cross-validation, MCC optimization, and threshold tuning to address class imbalance. Demonstrated that small, self-collected datasets can support meaningful prediction, as a proof of concept for resource-limited institutions.

Science Research Seminar

2025–2026

Selective lab-prep seminar — analyzing publications, passing a rigorous qualifying exam, and covering research ethics and statistics, plus a rigorous ML / Advanced Python course with hands-on projects.

English Peer Tutor

2025–2026

Chosen as one of the top writers in the grade to mentor younger students in grammar, structure, analysis, and creative writing.

03 SOFTWARE PROJECTS

Bloom

Solo · 2025

Shipped a full-stack habit-tracking app with Google auth, Supabase database, built-in AI coach, and a research-backed design, all living on Cloudflare Workers. Rejecting minimalism, the app is centered around the user's tree, where consistency translates into growth from a seedling into a beautiful tree.

Khromate

Solo · 2025

Built a site that allows users to generate royalty-free art from coded algorithms, alongside being able to download and post them to the in-site gallery.

04 ENGINEERING PROJECTS

Ocean Bathymetry & Wave Size

Solo · 2024

Built a working wave tank with interchangeable seafloor topographies to measure how slope, depth, and contour reshape waves.

Wing Shape & Aircraft Flight

Team · 2025

Engineered a modular sailplane swapping equal-area wings of varying aspect ratio; ran flight trials to isolate lift, stability, and glide.

05

LEADERSHIP & ACTIVITIES

Cross Country & Track

2025–Present

Year-round varsity athlete training 7 days/week, top-ranked in school. Won 2nd in the PSAL Sophomore 4×400m and 4th in the 800m in his first season. Team leader who runs workouts, mentors, and assists race-day logistics.

Interact Club of Chinatown — VP & Co-Founder

2025–Present

Co-founded Chinatown's Rotary youth chapter, the first of its kind in NYC. Built the board, membership, and comms from scratch, all while raising \$1,500 in year one and organizing volunteer events. Also runs socials and the club website.

06

COMMUNITY & SERVICE

Laru Beya Summer Volunteer

2024–Present

Nonprofit surf instruction and water safety for underserved youth.

Catholic Church Volunteer

2025–Present

Biweekly refugee clothing drives, multilingual book drives, property maintenance.

Photography Assistant

2024–Present

On-set production support, shooting/editing promo content, and prepping fine-art prints for gallery clients.

07

SKILLS

AI-Assisted Engineering

Claude Code

Agent Workflows

CLAUDE.md Specs

AI-Assisted Coding & Data Analysis

Full App Launching

Programming & Data

Python

pandas

Supervised Learning

Neural Networks

Data Pipelines

Game Development

Full-Stack & Version Control

Supabase

GitHub

Cloudflare

Render

Netlify

Lab Techniques

PCR

Gel Electrophoresis

Micropipetting

Distillation

Software

Excel

Word

PowerPoint

Logic Pro

Pixelmator Pro

Adobe Express

Affinity

Music Production

Recording Equipment

DAWs

Sampling

Piano

Languages

English (Fluent)

French (Fluent)